

Worksheet 2B: Linear Thinking — Data & Curves

Annotated Revision Guide

AIML Foundations Mathematics Dublin and Dún Laoghaire ETB

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Part A: Which Dataset Is Most Linear? (Problems 1–4)

Concept. A **linear relationship** means the data points cluster closely around a straight line. The **slope** of that line is the rate of change: how much y changes per unit increase in x . Real data is never perfectly linear, but linear models are powerful when the non-linear deviations are small relative to the overall trend.

Method. Visual check: does the data "zigzag" (scattered), follow a curve (diminishing returns or accelerating), or increase/decrease at a roughly constant rate (linear)? Rough slope: $m \approx \frac{\Delta y}{\Delta x}$ between first and last points.

Watch out. Diminishing returns (a flattening curve) can look linear at first. Check whether equal x -increments produce equal y -increments. If the y -gaps are shrinking, the relationship is concave, not linear.

Where this leads. The formal measure of linearity is the **correlation coefficient** r . Part E of this worksheet introduces r and its limitations.

Explore further. Tukey, *Exploratory Data Analysis* (1977) — the original guide to spotting patterns in data before fitting models.

1. Three datasets: Hours studied vs Exam Score.

- (a) **Most linear.** Dataset A: increases of roughly 5–7 points per hour throughout.
(b) **Most scattered.** Dataset B: large irregular jumps (50, 55, 70, 60, 85...).
(c) **Shows a curve.** Dataset C: fast gains early, then slowing (40, 45, 55, 70, 82, 88, 92, 94) — diminishing returns.

- (d) **Slope of A.** From (1, 45) to (8, 84): $m \approx \frac{84-45}{8-1} = \frac{39}{7} \approx 5.6$ points per hour.

Note: A slope of 5.6 means roughly a 6-point improvement in score for each additional hour of study, on average.

2. Temperature (°C) vs Ice Cream Sales (€).

- (a) **More linear.** Dataset P: roughly equal €60 increase per 5°C.
(b) **Slope direction.** Positive: higher temperature $\Rightarrow$ more sales.
(c) **Slope of P.** From (10, 120) to (35, 440): $m = \frac{320}{25} \approx 13$ € per degree.
(d) **Predict at 40°C.** $440 + 13 \times 5 \approx \text{€}505$.

3. Year vs Temperature Anomaly (°C above baseline).

- (a) **More consistent.** Dataset Y: steady 0.175°C per decade.
(b) **Slope of Y.** $\frac{0.90-0.20}{2020-1980} = \frac{0.70}{40} = 0.0175^\circ\text{C}/\text{year}$, i.e. 0.175°C per decade.
(c) **Predict 2050.** $0.90 + 0.175 \times 3 = 0.90 + 0.525 \approx 1.4^\circ\text{C}$.

(d) **Real data non-linear.** Natural variability (volcanic eruptions, El Niño, ocean cycles) adds noise around the trend.

Look ahead: The Paris Agreement target is 1.5°C above pre-industrial. Dataset Y's linear extrapolation reaches this around 2053. Real data (Dataset X) is noisy but broadly consistent with this trend.

4. Training Epochs vs Model Accuracy.

(a) **Diminishing returns.** Dataset M: gains of 12, 9, 6, 4, 2 — shrinking increments.

(b) **Perfectly linear.** Dataset N: exactly 10% per 10 epochs.

(c) **N is unrealistic.** Accuracy is bounded above by 1.0; Dataset N reaches 1.05 at epoch 60, which is impossible.

(d) **Steepest slope.** Between epochs 10 and 20 (gain of 0.12).

Part B: Positive, Negative, or Zero Slope? (Problems 5–6)

Concept. Slope direction tells you whether y increases or decreases as x grows. A **positive slope** means a direct relationship; **negative** means inverse; **zero** means no relationship (or no *linear* one).

Method. For each scenario, ask: "If x increases by 1, does y tend to go up, down, or stay the same?" Real-world knowledge is all that is required.

Watch out. Correlation is not causation, and zero *linear* correlation does not mean no relationship — a U-shaped curve has $r \approx 0$ but a strong pattern.

Where this leads. Slope values in machine learning: the gradient $\partial L / \partial w$ is the "slope" of the loss surface with respect to each weight. Gradient descent follows the negative slope.

Explore further. Freedman, Pisani & Purves, *Statistics* (4th ed.) — the clearest introduction to correlation and regression.

5. Positive, negative, or zero slope for each scenario?

(a) **Price vs quantity demanded. Negative** (law of demand: higher price, lower quantity).

(b) **Experience vs salary. Positive.**

(c) **Distance driven vs fuel remaining. Negative.**

(d) **Shoe size vs intelligence. Zero** (no relationship).

(e) **Altitude vs air pressure. Negative.**

(f) **Sleep vs alertness. Positive** (up to about 8 hours).

6. Match slopes -2 , -0.5 , 0 , 0.5 , 2 to descriptions.

- (a) y decreases by 2 per unit x . Slope = -2 .
- (b) y increases by 1 per 2 units x . Slope = 0.5 .
- (c) x has no effect on y . Slope = 0 .
- (d) y increases by 2 per unit x . Slope = 2 .
- (e) y decreases by 1 per 2 units x . Slope = -0.5 .

Part C: Line of Best Fit and Residuals (Problems 7–10)

Concept. The **line of best fit** (least-squares regression line) minimises the sum of squared residuals $\sum(y_i - \hat{y}_i)^2$. A **residual** is the vertical distance between a data point and the line: positive if the point is above, negative if below.

Method. Eyeballing: draw a line so roughly half the points are above and half below, with small, balanced vertical gaps. Residual for point (x_i, y_i) : compute $\hat{y}_i = mx_i + b$, then residual = $y_i - \hat{y}_i$.

Watch out. Squaring residuals before summing (a) makes all terms positive and (b) penalises large outliers more heavily. Without squaring, $+5$ and -5 residuals cancel to zero, suggesting a perfect fit when one is far from the truth.

Where this leads. The sum of squared residuals is the **mean squared error (MSE)** loss used in linear regression training. Minimising MSE with calculus gives the normal equations.

Explore further. The least-squares method was independently invented by Legendre (1805) and Gauss (1809) — and they argued bitterly about priority. See Stigler, *The History of Statistics* (1986).

7. Points $(1, 2), (2, 3), (3, 5), (4, 4), (5, 6), (6, 7), (7, 8)$: estimate line of best fit.

Trend. Generally increasing; the points are close to a line of slope ≈ 1 .

y -intercept. At $x = 0$ the line appears to cross near $y \approx 1$.

Slope. Rise from $x = 1$ to $x = 7$: about 6 units. Run: 6. Slope ≈ 1 .

Equation. $y \approx x + 1$.

8. Residuals for $y = x + 1$ using the points above.

- (a) **Point** $(1, 2)$. Predicted: $1 + 1 = 2$. Residual = $2 - 2 = 0$.
- (b) **Point** $(3, 5)$. Predicted: $3 + 1 = 4$. Residual = $5 - 4 = +1$.
- (c) **Point** $(4, 4)$. Predicted: $4 + 1 = 5$. Residual = $4 - 5 = -1$.
- (d) **Good fit?** Yes: residuals are small and balanced (positive and negative). $y = x + 1$ is a reasonable fit.

9. Student A: $y = 2x + 1$; Student B: $y = 1.8x + 1.5$.

- (a) **Smaller residuals.** Student B: maximum |residual| ≈ 0.8 vs Student A's 2.
- (b) **Better fit.** Student B.

(c) **Measuring total error.** Sum of **squared** residuals: $\sum (y_i - \hat{y}_i)^2$.

10. Why square the residuals instead of summing them directly?

Answer. Squaring makes all terms non-negative. Without squaring, residuals of $+5$ and -5 cancel to 0, falsely suggesting zero total error. Squaring also penalises large errors more than small ones, which is usually desirable.

Note: This is the key insight behind least-squares: the "loss" is the sum of squares, not sum of absolute values. Taking the absolute value is also valid (L1 loss) but less mathematically convenient because $|x|$ is not differentiable at 0.

Part D: Slope on Curves — Tangent Line Intuition (Problems 11–20)

Concept. On a curve, slope *changes* from point to point. At any specific point, the slope equals the slope of the **tangent line** — the unique line that "just touches" the curve there. Key signs: slope positive $\Rightarrow$ curve rising; negative $\Rightarrow$ falling; zero $\Rightarrow$ flat (local max or min).

Method. For $y = x^2$: the tangent slope at $x = a$ is $2a$ (from calculus). At $x = 0$, slope = 0 (bottom of parabola). For $y = \sin x$: slope at peaks/troughs is 0; slope at 0 and π is ± 1 . For exponentials: slope is always positive, and increases as x grows.

Watch out. Do not confuse the *value* of the function with its *slope*. $y = x^2$ has $y = 4$ at $x = 2$, but slope = 4 there too (coincidentally). At $x = 3$: $y = 9$ but slope = 6.

Where this leads. The slope of the tangent line *is* the derivative. Worksheet 4 is entirely about computing derivatives and using them. Every problem in Part D is an informal derivative calculation.

Explore further. 3Blue1Brown, *Essence of Calculus* (YouTube, 2017) — Chapter 2 visualises tangent lines and derivatives beautifully.

11. Parabola $y = x^2$.

- (a) **At $x = 0$.** Bottom of curve: tangent is **flat** (slope = 0).
- (b) **At $x = 2$ (right side).** Curve rising: slope **positive**.
- (c) **At $x = -2$ (left side).** Curve falling left-to-right: slope **negative**.
- (d) **Slope = 0.** Only at the vertex: $x = 0$.

12. Upside-down parabola $y = -x^2 + 4$.

- (a) **At $x = 0$ (peak).** Slope = 0 (tangent horizontal at maximum).
- (b) **At $x = 1$.** Curve descending to right: slope **negative**.
- (c) **At $x = -1$.** Curve ascending to reach peak: slope **positive**.

13. Exponential growth $y = 2^x$.

- (a) **Slope ever negative?** No: $2^x > 0$ always and always increasing.
- (b) **Slope ever zero?** No: the curve never flattens.
- (c) **As x increases.** Slope gets **steeper** (exponential growth accelerates).
- (d) **Slope smallest.** As $x \rightarrow -\infty$ (far left): approaches 0 but never reaches it.

14. Exponential decay $y = (1/2)^x$.

- (a) **Slope sign. Negative** everywhere: curve always decreasing.
- (b) **Steepest.** At far left (small x), where the curve drops most steeply.
- (c) **As $x \rightarrow \infty$.** Slope approaches 0: curve nearly flat.

15. Sine wave $y = \sin x$.

- (a) **At $x = 0$.** Curve rising (from 0 upward): slope **positive**.
- (b) **At $x = \pi/2$ (peak).** Slope = 0 (tangent horizontal).
- (c) **At $x = \pi$.** Curve descending: slope **negative**.
- (d) **At $x = 3\pi/2$ (trough).** Slope = 0 (tangent horizontal).
- (e) **Most positive/negative.** Most positive at $x = 0, 2\pi$ (crossing zero going up). Most negative at $x = \pi$ (crossing zero going down).

Note: The derivative of $\sin x$ is $\cos x$. Verify: $\cos(0) = 1$ (positive), $\cos(\pi/2) = 0$ (flat), $\cos(\pi) = -1$ (most negative). Everything here is consistent.

16. Cubic $y = x^3$.

- (a) **At $x = 0$.** The curve briefly levels off: slope = 0.
- (b) **Ever decreasing?** No: $y = x^3$ is always non-decreasing (slope = $3x^2 \geq 0$).
- (c) **Constant?** No: slope changes at every point (except the inflection at $x = 0$).

17. Sketch a curve for each description.

- (a) Slope always positive but decreasing: e.g. $y = \sqrt{x}$ or $y = \ln x$. Starts steep, flattens.
- (b) Slope always negative but increasing toward zero: e.g. $y = -e^{-x}$ or $y = -1/x$. Descends steeply at left, levels off at right.
- (c) Slope positive, then zero, then negative: e.g. $y = -x^2 + 4$ (or any downward parabola). Rises to a peak, then falls.

Note: These are sketching exercises. Any curve with the described slope behaviour is acceptable.

18. A car's position vs time (curve rising steeply then flattening).

- (a) **Moving forward?** Yes: position is increasing.
- (b) **Speeding up or slowing?** Slowing down: the curve is concave (slope decreasing).
- (c) **Moving fastest.** At the beginning, where the tangent is steepest.
- (d) **Horizontal tangent.** The car is momentarily stationary.

19. Ball thrown upward: height vs time (parabola opening downward).

- (a) **Left side.** Ball rising: slope **positive** (height increasing).
- (b) **At peak.** Slope = 0: momentarily neither rising nor falling.
- (c) **Right side.** Ball falling: slope **negative**.
- (d) **Physical meaning.** Slope = **velocity** (metres per second). Positive means upward; negative means downward.

Look ahead: This is exactly the setup for Worksheet 3C Part A. The height function $h(t) = -5t^2 + v_0t + h_0$ is a downward parabola whose slope (the derivative) gives instantaneous velocity.

20. Sketch a curve for each slope description.

- (a) **Zero at $x = 0$, positive for $x > 0$, negative for $x < 0$.** Example: $y = x^3$. Flat at origin, rising right, falling left.
- (b) **Always positive and always increasing.** Example: $y = e^x$. Always rising, accelerating.
- (c) **Positive then zero then negative then zero then positive.** Example: $y = \sin x$. A full oscillation.

Part E: Lines and Data Science (Problems 21–24)

Concept. **Linear regression** finds the line $y = mx + b$ that minimises the total squared error. The **correlation coefficient** $r \in [-1, 1]$ measures how tightly data clusters around a line. $|r|$ near 1 means strong linear relationship; near 0 means weak or none.

Method. Prediction: substitute x into $\hat{y} = mx + b$. Residual: $e = y - \hat{y}$. Squared residual: e^2 . Matching r values: $r > 0.8$ is strong positive; $r < -0.8$ is strong negative; $|r| < 0.3$ is weak.

Watch out. **Correlation $\neq$ causation.** Ice cream sales and drowning deaths are positively correlated because both are caused by hot weather — a **confounding variable**. No direct link exists.

Where this leads. The gradient of the MSE loss with respect to slope m and intercept b can be set to zero to get the **normal equations**: $m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$, $b = \bar{y} - m\bar{x}$.

Explore further. Pearl & Mackenzie, *The Book of Why* (2018) — the definitive treatment of causality vs correlation.

21. Line $y = 2x + 3$.

- (a) **Predict** $x = 5$. $\hat{y} = 2(5) + 3 = 13$.
- (b) **Residual** (actual $y = 14$). $14 - 13 = +1$.
- (c) **Squared residual**. $1^2 = 1$.

22. Match r values to descriptions.

- $r = 0.95$. **Strong positive trend.**
- $r = -0.85$. **Strong negative trend.**
- $r = 0.12$. **Very weak relationship.**
- $r = -0.02$. **Almost no relationship.**

23. "Correlation does not imply causation." Explain each spurious correlation.

- (a) **Ice cream and drowning.** Both caused by hot weather (**confounding variable**: summer).
- (b) **Firefighters and damage.** Larger fires *require* more firefighters; fire size causes both.
- (c) **Shoe size and vocabulary in children.** Both caused by **age**: older children have larger feet and larger vocabularies.

24. House price model: $\text{price} = 150000 + 200 \times \text{sq_ft}$.

- (a) **Slope meaning.** Each additional square foot adds €200 to the predicted price.
- (b) **Intercept meaning.** A theoretical house with 0 sq ft costs €150,000 — represents land value and fixed costs.
- (c) **Is intercept realistic?** Not literally (no 0-sq-ft house), but it captures the baseline cost of land, permits, infrastructure.
- (d) **Predict 1500 sq ft.** $150000 + 200 \times 1500 = 150000 + 300000 = \text{€ } 450000$.